Business Intelligence Applications

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1. Introduction and User Manual

1.1 Business Scenario

The objective of this project is to develop a comprehensive, interactive Business Intelligence application for a global retail company. The company operates in multiple international markets, selling various products across different categories.

1.2 Proyect Scope and Objectives

The primary goal of this application is to provide management with deep insights into sales activities through data-driven reporting in Power BI. The solution is designed to handle raw sales data without external pre-processing, ensuring a fully automated ETL (Extract, Transform, Load) pipeline within the Power BI environment.

1.3 Market Selection

In accordance with the assignment requirements, this report analyses three specific geographic markets:

- Market 1: United States of America (USA).
- Market 2: Germany.
- Market 3: Japan

1.4 Target Audience

The dashboard is designed for monthly management reviews, allowing users to drill down from high-level country overviews to specific product categories, cities, and individual ZIP codes.

2. ETL Process (Extract, Transform, Load)

2.1 Data preprocessing

The extraction was from raw csv provided which was transactional data from the assigned countries ,as well as the master data table which contains information from all countries product types and all necesary info to for with, we loaded them into PowerBI without external modification.

2.2 Data Cleaning & Transformation

Once the raw data was loaded into Power Query, a standardized ETL framework was implemented to normalize disparate data structures and ensure seamless integration across the US, German, and Japanese datasets.

To fullfil with European and German regional settings, decimal points in the "Revenue" field were replaced with commas.Percentage-based financial measures were rounded to zero decimal places, while all other financial values were rounded to one decimal place, with appropriate data type assignments applied to ensure consistent formatting. The transformation process also involved a significant structural alignment: for the German dataset, headers were translated from "Anzahl," "Umsatz," and "Land" to "Units," "Revenue," and "Country," while the date fields across all datasets were standardized to "SalesDate."

In the US dataset, a custom "Country" column was added to facilitate global consolidation. Regarding geographical data, cleaning was applied to the ZIP code fields to remove non-visible characters and extra blank spaces through "Trim" and "Clean" functions.

For the Japanese dataset, where the "CityZip" field was renamed to "Zip," a specific transformation was implemented using character selection logic to preserve only numbers and hyphens, strictly maintaining the "XXX-XXXX" format. This step prevents confusion with other international postal formats and ensures the geographical identity of Japanese locations remains intact. By performing these ETL steps within Power BI, the entire preparation process remains fully automated and scalable for future data loads.

3. Data Modeling and Dimensional Detail

The model uses a star schema and a single central query as the data source. Instead of connecting multiple times to the "master data.xlsx" file, a query called **MasterSource** serves as the sole gateway for all dimensions. This design ensures data consistency across the model, significantly improves refresh performance by reducing file access to a single operation, and simplifies maintenance by allowing file path updates to be managed in one central location.

To facilitate intuitive navigation and "drill-down" analysis in the dashboard, the following hierarchies were implemented:

- **Time Hierarchy:** Year → Quarter → Month Name.
- **Product Hierarchy:** Category → Product Type.
- **Geography Hierarchy:** Country → State → City → Zip.

3.1 Dimension Tables (D-Tables)

D_Geo (Geography Dimension) After skipping initial metadata and promoting headers, a filtering logic was applied to remove null entries and restrict the dataset to the assigned regions. For the German dataset, string manipulation was used to strip administrative prefixes such as "Kreisfreie Stadt" and "Landkreis," and a conditional logic unified city name (e.g., consolidating “Halle” and “Offenbach” variations). In the Japanese dataset, the "District" field was promoted to serve as the primary "City" name, and the state "Toukyouto" was standardized to "Tokyo." For US records, city names were isolated by removing state abbreviations after the comma. ZIP codes across all regions were cleaned to include only numeric characters and hyphens. Finally, redundant columns were removed, a unique **GeoID** index was generated as a primary key, and the table was reordered for optimal model performance.

D_Manufacturer (Manufacturer Dimension) To resolve the horizontal orientation of the raw source, a structural pivot was executed via a transpose operation. This aligned "ManufacturerID" and "Manufacturer" names into a standard columnar format.

D_Product (Product Dimension) The product dimension was improved to capture more detailed attributes. Raw product strings were split using the "|" delimiter to separate the **Product Type** (e.g., style details) from the main product name. The "Category" field was stabilized using a "Fill Down" operation, and the "Manufacturer" prefix was removed from the product names for clarity. Financial normalization was also a key focus: the "USD" prefix was stripped from the Price field, and the values were converted from text to numeric decimal types. This conversion included replacing dots with commas to maintain compatibility with European regional settings, ensuring that the final table provides a consistent price-point attribute for every ProductID.

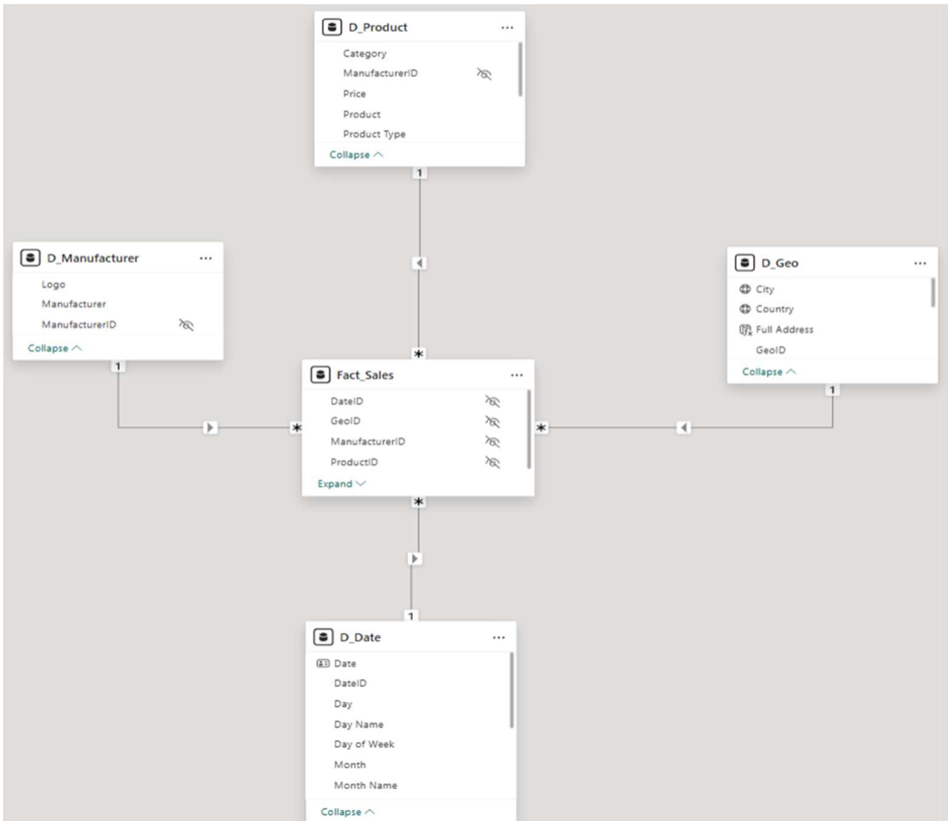
D_Date (Time Dimension) A dedicated calendar table was programmatically generated using Power Query M code, spanning a fixed range from January 1, 2014, to December 31, 2017. This table serves as the backbone for all Time Intelligence calculations. It establishes a comprehensive hierarchy including Year, Quarter (formatted as "Q#"), Month, and Day. To facilitate monthly-level reporting and user-friendly filtering, specific attributes such as "Month Name" and "Day Name" were localized in English. Finally, a **DateID** index was added to provide a unique integer key for efficient relationship mapping within the Star Schema.

3.2 The Fact Table (F_Sales)

To complete the star schema, a central **Fact Table** was created by consolidating the transactional data from the three assigned countries (USA, Germany, and Japan) into a single unified structure. This table serves as the primary engine for all quantitative analysis and measure calculations within the report.

Consolidation and Key Mapping: All country files were combined using a Table.Combine operation. Transactions were linked to their dimensions using **Zip**, **ProductID**, and **SalesDate** as keys. Left Outer Joins with **D_Product**, **D_Geo**, and **D_Date** replaced text and raw dates with numerical surrogate keys (**GeoID**, **DateID**, **ManufacturerID**) for a more efficient model.

Data Integrity and Final Optimization: ZIP codes in the Fact Table were cleaned using the same logic as in **D_Geo**, keeping only numbers and hyphens to avoid orphan records. The **Units** column was renamed **Sold Units** for clarity, all ID columns were cast to integers, and redundant columns like **SalesDate** and **Country** were removed to streamline the model and improve performance.



To ensure accurate geographical visualization, fields such as **City**, **Country**, **Full Address**, and **Zip** were explicitly assigned their respective **Data Categories**. Additionally, all technical keys used for relationships (DateID, GeoID, ManufacturerID, and ProductID) were hidden from the report view to provide a cleaner interface focused solely on business-relevant attributes.

4. Analytical Measures (DAX)

To transform raw transactional data into strategic business insights, a suite of measures was developed within the Fact_Sales table. These calculations focus on revenue optimization, historical growth trends, and performance monitoring.

4.1 Revenue and Gap Analysis

This framework evaluates sales efficiency by comparing actual results against theoretical potential:

- **Realized Revenue:** Total actual income generated from sales.
- **Possible Revenue:** The performance benchmark calculated as $Sold\ Units * Price$, representing maximum potential without discounts.
- **Revenue Opportunity Gap:** The absolute dollar loss between possible and realized revenue.

- **Opportunity Gap Target (10%):** A strategic threshold designed to absorb "normal" leakage (e.g., seasonal discounts or FX fluctuations). Exceeding this 10% limit flags critical revenue erosion.
- **Revenue Opportunity Gap %:** The efficiency ratio showing the percentage of potential revenue left on the table.

4.2 Historical Performance

To evaluate historical performance, specialized time-intelligence measures were implemented to smooth out seasonal volatility and reveal the true trajectory of the business:

Time-intelligence measures reveal the business trajectory by smoothing out seasonal volatility:

- **Revenue YTD:** Cumulative income from the start of the fiscal year to the current date.
- **Revenue PY & PY YTD:** Historical baselines for the same period and cumulative progress from the previous year.
- **Revenue YoY%:** The percentage growth rate comparing current performance to the prior year.
- **Revenue YoY YTD%:** Measures year-to-date cumulative growth against the previous year's pace to track long-term momentum.

5. Data Visualization and Interactive Reporting

The dashboard is structured into four specialized analytical views, accessible via a Navigation Menu. This modular design ensures that each page serves a specific business purpose while sharing a consistent header for global KPIs.

5.1 Global KPIs & Filters

The top section of every page remains static to provide immediate context and control:

- **KPI Cards:** Display high-level aggregates such as total revenue, units sold, and year-over-year growth percentages.
- **Indicator Card:** Visualizes the Revenue Opportunity Gap against its target, using conditional formatting to alert the user when the threshold is exceeded.
- **Slicers:** A set of four dropdown menus that allow for cross-filtering the entire report by Time, Manufacturer, Geography, and Product.

5.2 Executive Summary

This dashboard utilizes four key visualizations to provide a comprehensive and professional overview of business performance, offering a scannable layout designed to detect patterns and evaluate growth at a glance. A **multi-line chart** facilitates the comparison of trends and volatility across the three selected countries (USA, Germany, and Japan) over years and quarters , while a **cumulative area chart** provides an intuitive view of steady progress by comparing **Revenue YTD** against the **Revenue Previous Year YTD** baseline. To analyze efficiency, a **scatter plot** (Revenue Efficiency Analysis) compares **Units Sold** against the **Revenue Opportunity Gap %**; this normalization allows management to identify underperforming areas regardless of their size, flagging any location that exceeds the 10% tolerance threshold. Finally, a **lollipop chart** leverages the **Geographical Hierarchy**, allowing users to **drill down** from Countrt, State to City and ZIP code levels to analyze realized revenue and simplify the ranking of specific data points

5.3 Financial View (Structural Analysis)

This view is designed to support both auditing and growth analysis:

- **Waterfall Chart:** A variance visualization that explains revenue changes across years, quarters, and months by highlighting the positive and negative contributions of each manufacturer.
- **Matrix Table:** A structured data grid that presents the full set of DAX measures across the Date hierarchy, enabling side-by-side comparison of realized revenue, possible revenue, revenue gap, YoY and YoY%, prior-year YTD revenue and YTD%, as well as YoY YTD revenue and YoY YTD%.

5.4 Sales View (Volume and Distribution)

This section focuses on market share and product performance:

- **Donut Chart:** Illustrates the distribution of sold units by country, providing a clear view of market dominance.
- **Clustered Bar Charts:** Compare current performance against the previous year's baseline, broken down by Manufacturer, Product Type, and Category.
- **Line Chart:** Visualizes trends in unit sales and revenue over time, with drill-down capabilities to explore yearly, quarterly, and monthly patterns and identify seasonal peaks.

5.5 Geographical and Prescriptive View (Root Cause Analysis)

- **ArcGIS Heat Map:** A spatial visualization that highlights geographic sales hotspots based on unit sales density, helping identify high- and low-performing regions at a glance.
- **Decomposition Tree:** An AI-powered visual that enables users to break down total units sold into their contributing factors. The analysis follows a multi-level hierarchy—from Manufacturer to Category, Product Type, and Product ID—supporting root-cause analysis and revealing the key drivers behind performance at different levels of granularity.
- **Line Chart:** Monitors daily performance across states and weekdays, combining trend and comparison views to uncover operational patterns and temporal effects.
- **Clustered Column Chart:** Enables users to compare cities by units sold across categories, highlighting the top-performing cities within each category.

6. Content-based questions

Which city outside of the USA has had the highest realized revenue in 2017 and how much has that changed compared to 2016 (absolute and %)? .

The city outside of the USA with the highest realized revenue in 2017 is Kanazawashi. (Bookmark 1)

- Realized revenue in 2017: 5.2 million
- Change compared to 2016 (absolute): +120,097

Change compared to 2016 (percentage): +2.36%. (Bookmark 2)

2. What is the cumulated realized revenue in this city from January 2017 until May 2017?

The cumulated realized revenue in Kanazawashi from January 2017 to May 2017 amounts to 1.172.287. (Bookmark 3)

3. What would have been the “possible revenue” (=quantity * price) in this city?

The possible revenue in Kanazawashi, calculated as quantity multiplied by price, would have been 22.160.813. (Bookmark 4)

4. Which are the top three categories bought in this city? Create a report that allows to drill down to products interactively, too.

The top three categories purchased in this city are:

- Urban
- Rural
- Youth

An interactive report was created to allow drill-down from categories to individual products.

(Bookmark 5)

5. Which ZIP code area has had the largest revenue in this city in 2017?

The ZIP code area with the highest realized revenue in Kanazawashi in 2017 is 920-0952, with a total revenue of 488.8K. (Bookmark 6)

6. Analyze the change of realized revenues between 2016 and 2017 in this city with a waterfall visual using a breakdown based on manufacturers.

Between 2016 and 2017, revenue increased mainly because of Currus (+$0.12M) and Pirum (+$0.08M). Specifically, Currus grew by 22.66%. These gains were reduced by losses from Abbas (-$0.02M) and Natura (-$0.13M), but the overall results remained higher than the previous year. (Bookmark 7)

7. Use an additional AppSource visual (i.e. a non-standard visual) of your choice (see above) and apply it to this city

An additional non-standard AppSource visual was applied to the analysis of this city and is already integrated into the report.